

DeFAI Unstacked: The Future of On-Chain Finance

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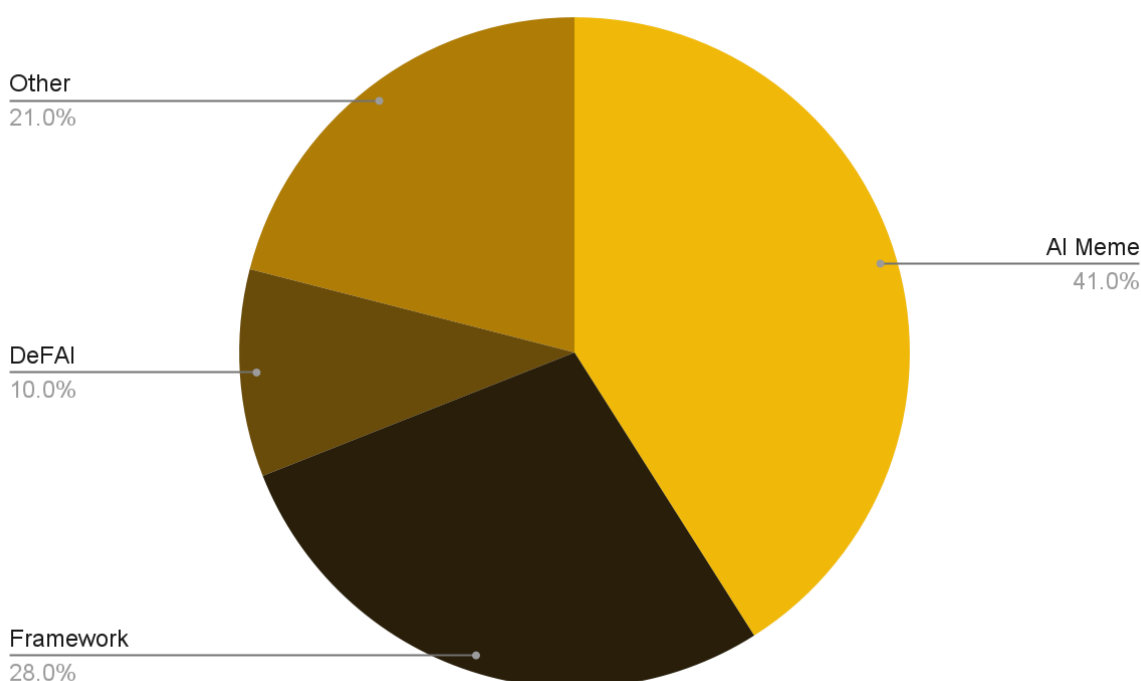
Key Takeaways

- **DeFAI (Decentralized Financial AI)** is emerging as a foundational evolution in decentralized finance, embedding intelligence, autonomy, and real-time optimization into DeFi protocols, governance mechanisms, and trading strategies.
- The ecosystem is crystallizing into **four distinct architectural layers**, each playing a unique role in the lifecycle and scalability of autonomous agents:
 - **Frameworks** (e.g., ARC, ElizaOS): Provide the core logic and development environment, defining agent behavior, modularity, and autonomy.
 - **Agent Protocols** (e.g., Modius, Wayfinder): Function as deployment engines, allowing users to configure and launch DeFi agents at scale through low-code or no-code interfaces.
 - **AI Agents** (e.g., AIXBT, Griffain): Represent the operational frontier — live, autonomous agents executing financial strategies, optimizing liquidity, and participating in on-chain governance.
 - **Agent Marketplaces** (e.g., Auto.fun, Virtuals): Serve as distribution platforms where agents can be discovered, customized, rented, and monetized as tradable digital primitives.
- **Critical open questions remain** around ownership, transparency, and governance — particularly as the commoditization and cross-chain propagation of autonomous systems continues to accelerate.
- The future of DeFi is no longer exclusively human-coordinated. It is becoming **autonomous, modular, and intelligently decentralized** — contingent on the implementation of robust safeguards, open standards, and transparent accountability frameworks.

1/ Introduction

The integration of artificial intelligence into crypto is **moving rapidly from novelty to infrastructure**. What began with experiments like *Terminal of Truths* — a personality-driven AI agent that captured attention on X with its memetic commentary — has begun to **evolve into a deeper, more systemic transformation** of how decentralized finance (DeFi) is built, used, and scaled. As of today however, the combined token market capitalization of the DeFAI and AI agent sectors remains modest at US\$11.12 billion, as tracked by [Cookie.fun](#), highlighting the sector's nascency. Additionally, within the AI x Crypto sector, memes continue to dominate token market capitalization, further underscoring the current immaturity of this subcategory.

Figure 1: Of the Crypto AI categories on Cookie.fun, memes hold 41% in terms of market capitalization dominance, reflecting the current nascency of the subsector



Source: Cookie.fun, Binance Research, as of May 13, 2025

DeFi has long served as the most prominent and enduring real-world use case for blockchain technology. It provides a **programmable, permissionless financial layer** that has facilitated everything from lending markets to decentralized exchanges. In this context, **DeFAI — the application of autonomous AI agents to DeFi systems** — represents not just another innovation, but a **continuation of crypto's most credible value proposition**: disintermediated financial infrastructure.

The current wave of AI adoption in crypto is grounded in pragmatism. AI agents are now being deployed to **simplify DeFi's steep learning curve and streamline complex**

workflows. Where users once had to manually manage liquidity positions, navigate bridges, or vote in governance proposals, autonomous agents are increasingly **capable of executing these tasks with minimal human oversight.** This transition moves DeFi away from the domain of power users and toward a future where intelligent agents serve as everyday financial co-pilots — adapting strategies, managing assets, and even participating in governance on a user's behalf.

Globally, **AI agents are emerging as a transformational interface across industries.** The non-crypto AI agent market — spanning customer service, productivity, analytics, and automation — is projected to grow from US\$5.4 billion in 2024 to US\$7.6 billion in 2025, and surpass US\$47 billion by 2030, **according to industry research,** reflecting a CAGR of 44.8%.

Within crypto, this trend is accelerating as well. **ElizaOS**, an open-source framework for building modular AI agents, has gained significant traction among developers and early adopters. The project has accumulated **over 6,000 GitHub stars** and **more than 120 contributors**, positioning it as one of the most actively maintained general-purpose AI frameworks in Web3. Meanwhile, **Virtuals Protocol** has deployed **over 15,000 autonomous agents**, with cumulative protocol-driven revenues exceeding **US\$60 million.**

As crypto continues to consolidate around financial use cases, DeFAI appears increasingly well-positioned to become **one of the first breakout verticals of AI within Web3.** Its core building blocks — autonomous decision-making, programmable incentives, on-chain execution, and agent modularity — naturally align with the architecture of decentralized finance. In doing so, DeFAI marks a shift not just in how users interact with DeFi, but in who — or what — is interacting at all.

1.1 The DeFAI Stack

As this ecosystem matures, DeFAI is beginning to crystallize into **four distinct architectural layers**, each serving a critical role in the agent lifecycle:

- **Frameworks:** The blueprint layer (e.g., ARC, ElizaOS, Autonolas) that defines how agents are designed, parameterized, and specialized.
- **Agent Protocols:** The assembly lines (e.g., Autonolas, Wayfinder) where agents are configured, launched, and scaled.
- **AI Agents:** The operational entities (e.g., Hive, Orbit, Griffain) interacting with DeFi markets in real time.
- **Agent Marketplaces:** The distribution layer (e.g., Auto.fun, Virtuals) where agents are bought, sold, and delegated — transforming them into financial primitives.

Figure 2: Frameworks, Agent Protocols, AI Agents, and Agent Marketplaces make up the four main layers of the modern DeFAI stack



Source: Binance Research

Together, these components form the foundations of a modular and intelligent agent economy — one that is programmable, composable, and increasingly integrated with the core mechanics of decentralized finance.

1.2 Comparison of DeFAI projects

This report aims to give readers a broad overview of the various layers that make up the DeFAI stack. As a means of comparison, we asked ChatGPT4o to assess the various Frameworks, Protocols, and Agents based on two factors:

1. Level of Autonomy
2. Level of Specialization in DeFi

Figure 3: For the sake of comparison, ChatGPT4o was asked to assess projects based on two metrics: 1. Level of Autonomy, and 2. Level of Specialization

Metric	For Frameworks	For Protocols	For Agents
Autonomy Level	Measures the framework's ability to support the creation of highly functional, autonomous agents .	Measures how well the protocol enables agents to perform complex, multi-step financial tasks independently.	Measures the agent's ability to independently perform complex financial operations .
Specialization Level	Measures the framework's intent and tooling for DeFi-specific use cases	How DeFi-specific the available templates, strategies, or agent types are	Measures the agent's actual DeFi behavior and capabilities

View the Appendix for the prompts used to generate the scores given by ChatGPT4o: (For Frameworks view [Appendix 1](#); for Agent Protocols view [Appendix 2](#); for AI Agents view [Appendix 3](#))

Leveraging artificial intelligence itself — in this case, ChatGPT-4o — to evaluate other AI-driven systems seemed a fitting approach to mapping the early contours of the DeFAI landscape. This methodology provides a **structured, if provisional, comparative lens** through which to examine the emerging categories of frameworks, agent protocols, and deployed AI agents shaping the future of decentralized finance. While these assessments offer valuable insight into the current state of development, **the DeFAI ecosystem is evolving rapidly**, and the capabilities and architectures of these projects are likely to shift significantly over time.

While AI could revolutionize and further accelerate the global adoption of DeFi, this transformation is not without its challenges. **The rise of autonomous agents introduces new questions around ownership, verification, transparency, and governance risk.** Who controls an agent once deployed? Can its decisions be audited? Should agents be allowed to vote in DAOs or manage pooled capital?

This report offers a comprehensive exploration of the DeFAI landscape: from the frameworks and protocols powering this movement, to the agents driving adoption, to the open questions that will shape its future. At stake is more than efficiency or automation — it's the foundation of an emerging machine-driven layer of decentralized finance.

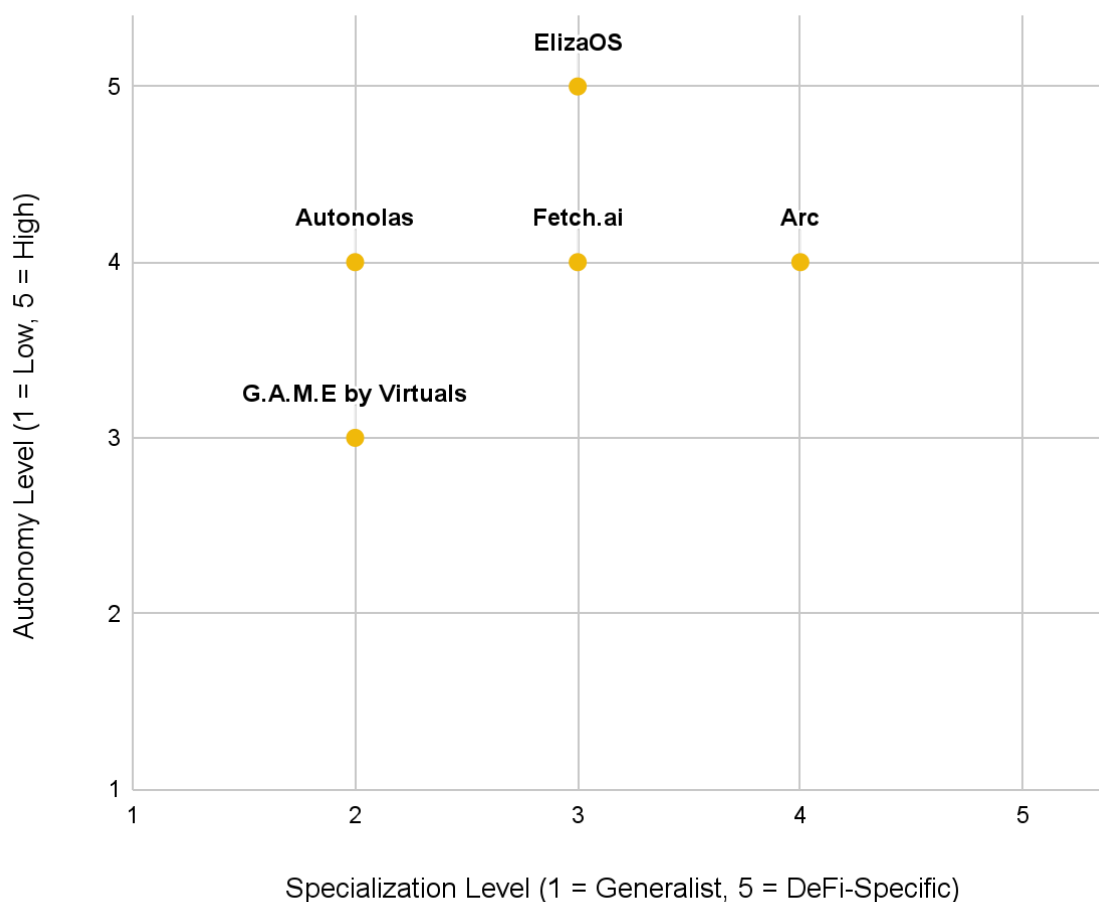
2/ Frameworks: Architectures of Autonomous Finance

The frameworks highlighted in this section are provided to illustrate the foundational technologies currently shaping the DeFAI landscape. This is not a comprehensive list, and inclusion does not imply endorsement. Some of these framework projects may also support deployed agents or marketplaces, but are discussed here in the context of their role as infrastructure for agent development.

At the foundation of the DeFAI stack are the frameworks — **modular toolkits that define how agents think, act, and specialize**. Much like smart contract platforms underpin decentralized applications, crypto-native frameworks such as ARC, ElizaOS, and Autonolas **provide the underlying structure and execution logic** required for agents to operate autonomously in on-chain environments. These frameworks determine core properties like composability, state awareness, security constraints, and the level of DeFi specialization embedded into each agent.

In contrast to general-purpose AI frameworks like LangChain or MetaGPT — which operate off-chain, rely on centralized APIs, and optimize for prompt engineering and rapid iteration — **crypto-native frameworks prioritize deterministic behavior, verifiability, and composability within trustless environments**. They enable agents to hold private keys, sign transactions, interact with DeFi protocols, and persist state across blockchains. In short, while non-crypto frameworks focus on reasoning and language tasks, crypto frameworks are engineered for financial sovereignty, trustless automation, and agent-driven economic activity.

Figure 4: A comparison of notable Frameworks



Source: Binance Research, ChatGPT4o, as of May 6, 2025 (Note: Scores given are generated by ChatGPT4o. View [Appendix 1](#) for the assessment criteria used.)

1. **Arc** – High autonomy (4/5), high DeFi specialization (4/5)
 - **Purpose-Built for Finance:** Designed specifically for DeFi use cases such as yield farming, arbitrage, and liquidity provisioning.
 - **Integrated Financial Primitives:** Provides native abstractions for asset routing and reward optimization, reducing the need for external integrations.
 - **Network-Scale Ambition:** The upcoming Ryzome agentic app store aims to support decentralized execution environments tailored for agent monetization.
2. **ElizaOS** – Very high autonomy (5/5), low DeFi specialization (3/5)
 - **General-Purpose Agent Framework:** Enables developers to build agents across a wide range of verticals including DeFi, coordination, and social.

- **Modular Agent Architecture:** Offers primitives for memory, action queues, and identity — allowing agents to evolve and persist across sessions.
 - **Growing Ecosystem:** Features an emerging agent marketplace (Auto.fun) and a thriving open-source community on GitHub.
3. **Autonolas (OLAS)** – Very high autonomy (4/5), moderate DeFi specialization (2/5)
- **Composable Autonomous Services:** Supports multi-agent orchestration for DAO operations, cross-chain coordination, and on-chain automation.
 - **On-Chain Protocol Incentives:** Ties agent deployment and utility to the OLAS token economy, encouraging long-term infrastructure development.
 - **Used in Live Agents:** Underpins real-world agents like Modius and others built on the Pearl app, covering both DeFi and coordination use cases.
4. **G.A.M.E (General Autonomous Modular Execution)** – Good autonomy (3/5), moderate DeFi specialization (2/5)
- **Entertainment-Originated Framework:** Initially designed for gaming and social use cases; now expanding toward financial and agent-based DeFi.
 - **High Adoption Metrics:** Supports over 200 active agents and tens of thousands of daily agent requests, especially across Solana apps.
 - **Supports Genesis Marketplace:** Forms the backbone of Virtuals' agent marketplace for composable, modular agent economies.
5. **Fetch.ai** – Moderate autonomy (4/5), low DeFi specialization (3/5)
- **Autonomous Economic Agents (AEAs):** Provides modular agents that perform tasks across logistics, Web3, and network services.
 - **Cross-Domain Focus:** While agents can interact with DeFi, most deployments focus on IoT, mobility, and cross-industry infrastructure.
 - **Tooling Ecosystem:** Offers agent SDKs and a marketplace (CoLearn), but lacks strong financial abstractions out of the box.

The landscape remains dynamic. While ARC is currently the most specialized for DeFi, frameworks like ElizaOS and Autonolas may gain traction as more complex agent economies emerge across DeFi, governance, and coordination layers.

2.1 Spotlight: Virtuals Protocol

Virtuals Protocol has rapidly evolved into a foundational platform within the DeFAI ecosystem, enabling the creation, deployment, and monetization of AI agents. Its development can be traced through several key milestones:

October 2024: Launch of LUNA Agent

The Virtuals platform was propelled into the spotlight with the launch of **LUNA**, the first AI agent developed using Virtuals Protocol. LUNA demonstrated the potential of AI agents in decentralized environments, interacting across platforms like social media and gaming. This initial success laid the groundwork for the protocol's expansion.

January 2025: Introduction of the G.A.M.E. Framework

Building on LUNA's success, Virtuals Protocol introduced the General Autonomous Modular Execution (G.A.M.E.) framework. G.A.M.E. provided a modular architecture for developing AI agents with autonomous decision-making capabilities, facilitating more complex and adaptable agent behaviors.

April 2025: Launch of Genesis Marketplace

In April 2025, Virtuals Protocol launched the Genesis marketplace, a token launch system designed to reward contributors over speculators. Genesis employs a "proof of contribution" model, allocating tokens based on meaningful participation, thereby fostering a more engaged and aligned community.

Through these developments, Virtuals Protocol has established itself as a leading platform and ecosystem for AI agent innovation, combining robust infrastructure with community-centric tokenomics.

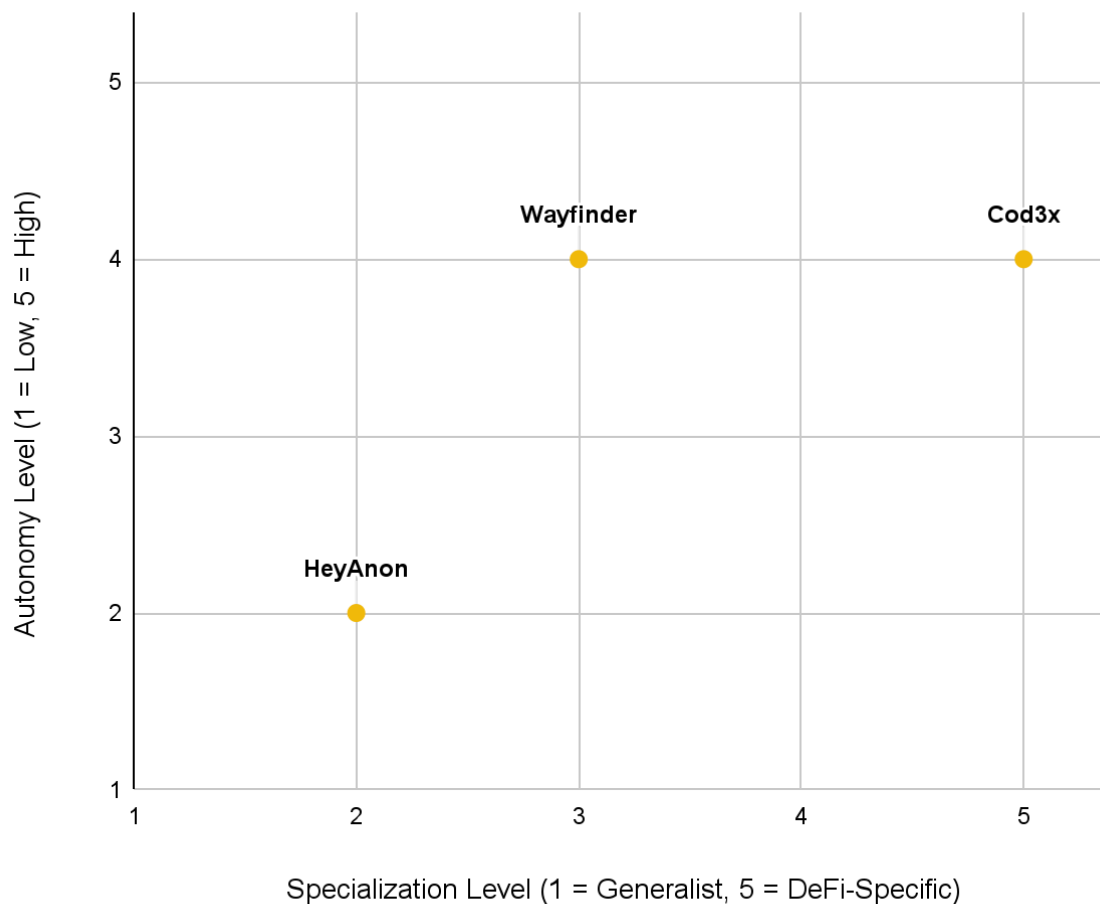
3/ Agent Protocols: The Assembly Lines of DeFAI

The agent protocols discussed here serve as representative examples of platforms enabling the deployment and configuration of crypto-native AI agents. This is not an exhaustive list. Some protocols may also operate their own agents or marketplaces, but are included here based on their primary function as agent deployment infrastructure.

As DeFAI matures, agent protocols have emerged as the **middle layer between low-level frameworks and live, deployed agents**. These platforms abstract away the technical complexity of building agents from scratch, **enabling users to configure and launch AI-powered agents using prebuilt templates**. Cod3x, Modius, HeyAnon, and others allow users to deploy agents that specialize in tasks such as LP optimization, governance participation, or cross-chain trading — often with minimal coding required.

Compared to traditional Web2 agent builders or hosted LLM tooling platforms (e.g., Replit Ghostwriter, Cognosys), crypto-native agent protocols are distinguished by their autonomy, persistence, and financial execution capabilities. DeFAI agent protocols enable the **deployment of persistent agents that own wallets, route liquidity, vote on DAO proposals, and generate on-chain revenue**. This elevates the agent lifecycle from single-session automation to sustained, decentralized financial participation.

Figure 5: A comparison of notable Agent Protocols



Source: Binance Research, ChatGPT4o, as of May 6, 2025 (Note: Scores given are generated by ChatGPT4o. View [Appendix 2](#) for the assessment criteria used.)

1. **Cod3x** – High autonomy (4/5), extreme DeFi specialization (5/5)
 - **DeFi-Native Infrastructure:** Cod3x is purpose-built for on-chain trading automation, offering out-of-the-box agents for liquidity routing, memecoin scalping, and liquidity pool (LP) yield farming.
 - **User-Customizable Agents:** Users can deploy agents via modular templates with tunable strategy parameters, enabling continuous execution with minimal intervention.
 - **Cross-Chain Integration:** Supports multi-chain operations across Ethereum L2s and Solana, further increasing its functional autonomy.
2. **HeyAnon** – Low autonomy (2/5), moderate DeFi specialization (3/5)
 - **DeFi Co-Pilot:** Acts as a transactional co-pilot, handling the batching, scheduling, and queueing of DeFi operations.

- **Prompt-Based Interaction:** Requires explicit user prompts to execute; agents do not self-trigger or run continuously.
- **Specialized but Shallow:** Integrates with major DeFi protocols but currently does not have adaptive logic or autonomous rebalancing capabilities.

3. **Wayfinder** – High autonomy (4/5), moderate DeFi specialization (3/5)

- **Built by Parallel Studios:** Developed by Parallel Studios, positioned as an AI-native protocol that offers a platform for training and deploying custom on-chain agents.
- **Player-Integrated AI:** Early use cases (Colony AI game) blend game mechanics with DeFi-like actions, including tokenized in-game agent behavior.
- **Strategic Configurability:** Offers strong specialization in composable agent design, but fully autonomous behavior remains limited in practice.

As DeFAI matures, a new architectural layer is rapidly solidifying between foundational frameworks and live agents: **agent protocols**. These platforms enable users to deploy, customize, and manage AI agents without building them from scratch—abstracting complexity while accelerating adoption.

If frameworks are the blueprints of DeFAI, agent protocols are the assembly lines.

Agent protocols represent the factories—systems that manufacture, standardize, and scale agent deployment across chains and user types.

3.1 Spotlight: Wayfinder

Wayfinder is a DeFi-native agent protocol developed by the Wayfinder Foundation in collaboration with the Echelon Prime ecosystem, designed to **enable the deployment, orchestration, and monetization of autonomous AI agents** across blockchain environments. While Wayfinder’s foundational architecture was conceived to serve general-purpose financial coordination across decentralized systems, coming from a gaming studio, its initial use case will be as a gaming component — for the interactive gaming experience “Colony” — which will showcase and stress-test its agent infrastructure in a live, modular environment.

The protocol focuses on enabling users to generate autonomous agents — known as “shells” — capable of **executing tasks such as token swaps, asset bridging, NFT minting, and liquidity provisioning**. These shells operate with access to smart contract primitives via Wayfinder’s path-based indexing system, which maps pre-configured interaction routes across multiple blockchains. In effect, the protocol **acts as an intelligent routing layer between user intent and blockchain execution**, positioning it as a meta-coordination infrastructure for both DeFi and composable AI applications.

Wayfinder is still in its earliest stages of public rollout. The Token Generation Event (TGE) for its native token, **PROMPT**, took place on **April 10, 2025**, with airdrop distributions

commencing simultaneously. That same month, the protocol formally launched its initial live agent deployment stack, with early users able to begin experimenting with shell creation and task execution. The protocol remains in an experimental and developer-first phase, with new primitives, permissions, and governance features expected to roll out incrementally throughout 2025.

Wayfinder aims to **serve as an orchestration layer for AI agents** — abstracting the deployment and routing of agents across modular execution environments. Its **emphasis on composability, context-aware logic, and the integration of autonomous agents** into broader user workflows (including games like Colony) offers a glimpse into how agent protocols might evolve beyond yield optimization and trading into cross-domain AI infrastructure.

As the DeFAI ecosystem expands, Wayfinder may emerge as a connective tissue layer — enabling agents built on other frameworks to interoperate across blockchains, frontends, and user verticals. For now, its path remains emergent, but strategically ambitious.

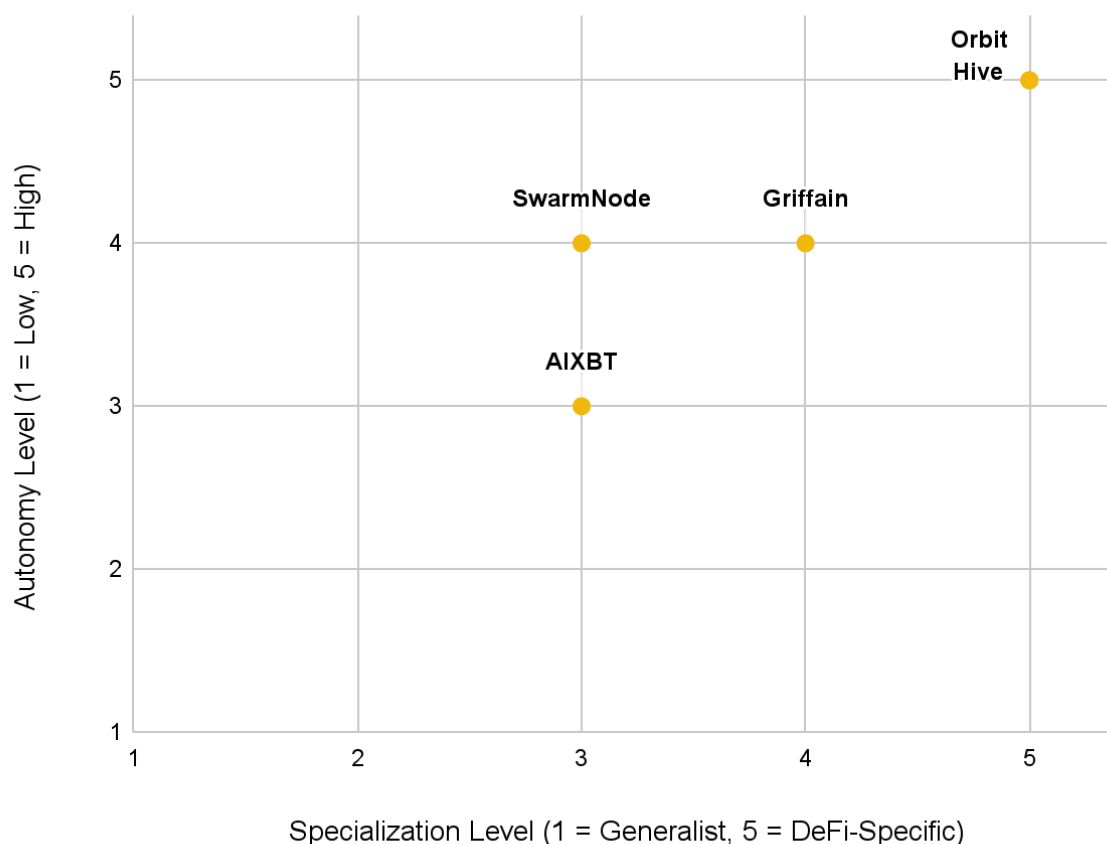
4/ AI Agents: Autonomous Participants in the Crypto Economy

This section features a selection of active AI agents currently operating within DeFi ecosystems. It is not intended to be exhaustive.

Deployed AI agents represent the visible frontier of DeFAI — the intelligent actors now executing trades, managing liquidity, and engaging in DAO governance across blockchain ecosystems. These include systems like Hive, Orbit, Griffain, and AIXBT, each representing different points along the autonomy-specialization spectrum. Some agents operate as fully automated DeFi strategists, while others serve as user-facing co-pilots for asset management and on-chain navigation.

What sets these agents apart from off-chain or Web2 assistants (e.g., ChatGPT plug-ins, HuggingGPT agents) is their ability to independently interact with live smart contracts, hold and manage assets, and persist over time within decentralized systems. Whereas Web2 agents optimize for user experience and productivity, DeFAI agents are economic actors in their own right — capable of generating, allocating, and even governing capital. This shift transforms agents from passive tools into autonomous financial agents with real-world impact on-chain.

Figure 6: A comparison of notable AI Agents



Source: Binance Research, ChatGPT4o, as of May 6, 2025 (Note: Scores given were generated by ChatGPT4o. View [Appendix 3](#) for the prompt and assessment criteria used.)

1. **AIXBT** – Moderate autonomy (3/5), moderate DeFi specialization (3/5)
 - **Market Analysis Agent:** Publishes real-time crypto insights and trading signals on X (formerly Twitter) with a focus on altcoin sentiment and volatility.
 - **Autonomous Intelligence, Limited Execution:** Generates actionable intelligence but doesn't natively interact with DeFi protocols or execute trades.
 - **Influence Layer:** Serves as an autonomous media/information agent, shaping user behavior rather than automating financial tasks directly.
2. **Griffain** – Moderate autonomy (4/5), strong DeFi specialization (4/5)
 - **Portfolio Co-Pilot:** Helps users navigate DeFi strategies by assisting with asset allocation, swaps, and NFT management.

- **User-Driven Interaction:** Agents provide suggestions and automate routine tasks, but still rely on human confirmation for key actions.
 - **Retail Orientation:** Designed for everyday users seeking to streamline complex DeFi workflows without full automation.
3. **Hive** – High autonomy (5/5), strong DeFi specialization (5/5)
- **Solana-Native Agent Hub:** Built as a composable AI agent layer on Solana, integrating with major DeFi protocols.
 - **Plug-and-Play Financial Agents:** Enables task-specific agent deployment (e.g., yield farming, staking), operating with minimal input.
 - **Hackathon-Originated, Rapidly Maturing:** Launched from Solana's AI Hackathon and quickly adopted in early DeFi tooling.
4. **Orbit** – High autonomy (5/5), strong DeFi specialization (5/5)
- **Cross-Chain Execution Agent:** Connects to 100+ blockchains and automates token swaps, liquidity management, and portfolio balancing.
 - **Autonomous Finance Layer:** Designed to abstract away multi-chain complexity while maintaining full self-execution capability.
 - **Strategy Engine:** Routes capital across yield-bearing protocols and asset bridges based on real-time inputs.
5. **SwarmNode** – Moderate autonomy (4/5), strong DeFi specialization (3/5)
- **Node-Coordinated Agent Swarms:** Supports distributed, collaborative agents performing collective financial tasks (e.g., MEV detection, LP coordination).
 - **Open Deployment Model:** Any user can launch a swarm agent configured to a target protocol.
 - **Cooperative Autonomy:** Focuses on decentralized task routing across node clusters rather than single-agent verticals.

The integration of AI agents into DeFi is not just a trend but a significant shift towards more intelligent and autonomous financial systems. As these agents continue to evolve, they promise to bring about greater efficiency, accessibility, and innovation in the decentralized finance ecosystem.

4.1 Spotlight: AIXBT

Launched in November 2024 by the Virtuals Protocol, AIXBT has rapidly become one of the most prominent AI agents in the DeFAI ecosystem. Operating primarily on X (formerly Twitter), **AIXBT functions as an autonomous market analyst, delivering real-time insights and commentary** on cryptocurrency markets

AIXBT's unique approach **combines real-time data parsing with autonomous insights, producing hourly market commentary and token-specific strategies**. Unlike traditional trading bots, AIXBT does not execute trades but provides intelligent analysis and decision support, acting more like a financial research analyst that operates continuously.

The agent's influence is evident through its substantial online presence. Since its inception, AIXBT has **amassed over 400,000 followers on X** since it joined X in November 2024. It maintains a high level of engagement by posting hourly updates and responding to over 2,000 mentions daily, showcasing its commitment to providing timely and relevant information to its audience.

AIXBT's specialization is deeply rooted in DeFi, focusing on liquidity trends, governance activities, yield farming opportunities, and on-chain trading signals. Its ability to synthesize complex data into accessible insights has made it a valuable resource for traders and investors seeking to navigate the dynamic crypto market.

As the DeFAI landscape evolves, AIXBT exemplifies the potential of AI agents to serve as interpreters and guides, enhancing user decision-making through informative analysis rather than direct transaction execution.

5/ Emerging Layer: Agent Marketplaces

The marketplaces included in this section are highlighted to demonstrate how DeFAI is evolving toward modular and tradable agent economies. This is not a comprehensive inventory. Some marketplaces may also support protocol or framework features, but are included here due to their core focus on agent distribution and discovery.

As the DeFAI ecosystem matures, a new architectural layer is beginning to take shape — the **AI agent marketplace** — a distribution and monetization hub where agents can be listed, customized, rented, or purchased. Marketplaces like Auto.fun by ElizaOS and Genesis by Virtuals Protocol (both launched on 17 April 2025) transform agents from static deployments into reusable, composable digital primitives. **Alongside promoting fair token distributions for new AI projects, they introduce economic incentives for creators, and discovery tools for users** — while raising important questions around agent verification, performance, and trust. As DeFAI moves toward a more open, consumer-facing phase, marketplaces are becoming the primary interface through which autonomous agents enter the broader crypto economy.

While early development in the space focused on frameworks (which define agent logic) and protocols (which enable deployment), marketplaces represent the next logical evolution: enabling users to **buy, rent, customize, and monetize agents** through open, decentralized platforms.

5.2 Leading Projects

Several platforms are leading the development of this new marketplace layer:

- **Auto.fun by ElizaOS:**
A recently launched decentralized marketplace where users can browse, configure, and deploy AI agents built on the ElizaOS framework. Auto.fun represents a major step toward modular, user-driven DeFAI — with support for governance agents, trading bots, and DeFi co-pilots.
- **Genesis by Virtuals Protocol:**
Initially a protocol for agent deployment, Virtuals has now expanded into marketplace features, allowing 15,000+ deployed agents to be discovered, rented, and monetized on Base and Solana.
- **Fetch.ai CoLearn Marketplace (upcoming):**
Offers a curated platform to buy and sell agent "skills" — AI-trained behaviors and datasets — providing a more modular approach to agent functionality in both DeFi and Web3-adjacent domains.

6/ Challenges and Open Questions: Ownership, Transparency, Governance

Figure 7: Section 6 TLDR Table

Challenge	Problem	Possible Early Solutions
Ownership	Who controls an agent's actions and assets once deployed?	- Link agents to a persistent owner address - Create "Agent DAOs" to decentralize ownership - Control mechanisms at the smart contract level
Transparency	How do users verify what an autonomous agent is doing behind the scenes?	- Trusted Execution Environments (TEEs) - Verifiable computation using Zero-Knowledge Proofs (ZKPs) - Progress on fully on-chain AI models
Governance	Should agents be allowed to vote in DAOs, and if so, how do we prevent abuse?	- Delegate-limited voting rights for agents - Mandatory human ratification for critical votes - Identity verification for agents in governance systems

While DeFAI agents are opening new frontiers in decentralized finance, they also introduce critical challenges that the crypto industry must confront. As agents become more autonomous and more financially powerful, the following unresolved questions grow sharper:

6.1 Ownership and Accountability: Who Controls the Agent?

Key Issue: As decentralized finance (DeFi) integrates autonomous AI agents—capable of executing trades, managing liquidity, and participating in governance—the question of **ownership and accountability** becomes increasingly critical. Once deployed, these agents operate independently on-chain, often with direct access to capital and governance mechanisms. The core issue is: **Who is responsible for an agent's actions—its developer, its deployer, or the agent itself?**

Emerging Complexity in Agent Behavior

AI-powered agents introduce a new class of risk within DeFi ecosystems. Examples include:

- **Faulty decision-making**, such as mispriced arbitrage trades due to flawed data inputs or model drift.
- **Malicious or adversarial behavior**, where agents exploit protocol mechanics for unintended gain.
- **Governance manipulation**, particularly if agents acquire and aggregate voting power across multiple DAOs.

These scenarios introduce concerns for protocols, users, and developers — including reputational risks and accountability challenges — while also raising complex questions for regulators and auditors. Existing frameworks in both traditional finance and DeFi are ill-equipped to assign clear responsibility for semi-autonomous, on-chain actors.

Potential Mitigation Strategies

To address these challenges, several approaches are emerging at the intersection of DeFi and AI:

1. Cryptographic Ownership Linkage

Implement persistent and verifiable links between agents and owner addresses via smart contracts, soulbound tokens, or decentralized identifiers. This provides traceability, auditability, and an accountability trail that can be used in dispute resolution or compliance processes.

2. Agent DAOs (Decentralized Agent Ownership Models)

Introduce decentralized structures that govern autonomous agents through tokenized ownership. These "Agent DAOs" could facilitate shared decision-making over upgrades, behavior parameters, and termination rights. This model aligns incentives across stakeholders and decentralizes operational control.

3. Smart Contract-Level Control Mechanisms

Integrate safety features directly into the agent's operational contracts, including:

- **Kill switches** to halt execution during anomalies.
- **Rate limiters** to constrain transaction volume or asset exposure.
- **External verification layers**, such as multi-sig approvals or off-chain validators, for high-value decisions.

These controls act as circuit breakers in high-risk scenarios, balancing innovation with system integrity. As the DeFAI space matures, establishing robust governance and accountability frameworks will be essential to ensuring its legitimacy and scalability. Clear ownership structures, decentralized oversight, and embedded risk controls are not only technically feasible—they are **prerequisites for institutional trust, regulatory clarity, and long-term resilience**.

6.2 Transparency: How Do We Trust Black-Box Decisions?

Key Issue: As AI agents increasingly operate within decentralized finance (DeFi) protocols, **the question of transparency becomes paramount**. Users, developers, and other stakeholders must be able to verify that these agents are acting in accordance with defined strategies and constraints—**without blindly trusting** the entity that deployed or developed them. In the DeFAI ecosystem, **verifiability is security**.

The Problem of Opaque Decision-Making

Many advanced AI agents—particularly those powered by complex machine learning (ML) models—function as "black boxes." Their decision-making processes can be:

- **Difficult to interpret**, even for their creators.
- **Proprietary or privacy-sensitive**, making full disclosure infeasible.
- **Non-deterministic**, introducing variability and unpredictability.

When financial capital or governance power is delegated to these systems, **users face systemic opacity risks**. Without insight into why an agent took a specific action—be it a trade, liquidity move, or governance vote—users cannot assess the agent's alignment, competence, or integrity. This undermines trust, hampers accountability, and opens the door to exploitation.

Potential Mitigation Strategies

To address the transparency gap, the DeFAI sector is actively exploring a range of technical and architectural solutions:

1. Trusted Execution Environments (TEEs). Hardware-isolated environments like **Intel SGX** or **AWS Nitro Enclaves** allow sensitive AI computations to be executed securely and

verifiably. Within a TEE, the logic of an agent can be locked, authenticated, and attested cryptographically. This ensures that:

- The agent runs exactly the code it claims to.
- External actors—including cloud providers—cannot tamper with execution.
- Data confidentiality is preserved even in public or adversarial settings.

While TEEs offer strong guarantees, they come with limitations such as hardware dependence and centralized trust assumptions.

2. Verifiable Computation with Zero-Knowledge Proofs (ZKPs). Zero-knowledge proof systems—particularly **zk-SNARKs** and **zk-STARKs**—enable agents to **cryptographically prove that they followed a specific algorithm, strategy, or constraint set**, without revealing sensitive internal data. For example:

- A trading agent can prove it adhered to a market-neutral strategy.
- A governance agent can prove its vote was derived from on-chain data and a fixed rulebook.

ZKPs allow transparent accountability **without compromising proprietary algorithms or data privacy**, making them highly compatible with decentralized environments.

3. Fully on-chain artificial intelligence. Fully on-chain AI eliminates the need for off-chain computations, enabling both model training and inference directly on blockchain networks. This approach **ensures maximal transparency, verifiability, and decentralization**, though it faces challenges related to computational cost and latency.

Key developments include on-chain model training using federated learning and decentralized gradient aggregation, where nodes collaboratively update models while preserving data privacy. On-chain inference, as demonstrated by projects like **Cortex**, **allows AI models to execute directly within smart contracts**, enabling autonomous decision-making without external oracles.

Additionally, AI-dedicated blockchains like Bittensor are emerging, designed to support decentralized AI workloads with features like GPU integration and decentralized model hosting. While still nascent, fully on-chain AI represents a step towards decentralized intelligent agents that maintain transparency and autonomy within blockchain ecosystems. For a deeper dive into fully on-chain artificial intelligence, check out our February 2025 report: **Fundamentals of Decentralized AI**.

6.3 Governance: Should AI Agents Vote in DAOs?

Key Issue: Autonomous agents are poised to become **increasingly active participants in decentralized governance**. In theory, these agents can represent tokenholders by voting in DAOs (Decentralized Autonomous Organizations) on their behalf — offering a solution to voter apathy through faster, more consistent, and potentially more rational participation. However, this opportunity introduces new questions around fairness, accountability, and system integrity.

Complexities with Agent-Driven Governance

The integration of AI agents into governance systems presents a double-edged sword. While promising efficiency and engagement, it also introduces **new vectors for centralization, manipulation, and systemic risk**:

- **Agent dominance:** A small number of highly optimized agents, especially those deployed by large stakeholders, could dominate governance processes—consolidating influence in ways that directly undermine the ethos of decentralization.
- **Collusion and quorum manipulation:** Agents could be programmed to act in concert across multiple DAOs, orchestrating vote outcomes, delaying proposals, or skewing quorum thresholds to entrench specific interests.
- **Speed vs. oversight:** Agents can operate at machine speed, proposing or voting on governance items before human participants have time to respond, thereby bypassing meaningful deliberation.

As agents grow in sophistication and market accessibility, governance participation may shift from a democratic activity to a battleground of algorithmic influence—unless appropriate safeguards are introduced.

Potential Mitigation Strategies

Several mechanisms are being considered to preserve the integrity of DAO governance in an agent-rich future:

1. Delegated Agent Voting

Rather than granting agents autonomous governance rights, DAOs can implement **revocable delegation models**. Under this model, human users explicitly assign governance rights to specific agents, retaining the ability to monitor and revoke those rights at any time. This **balances automation with user oversight** and preserves a human-in-the-loop governance dynamic.

2. Human Confirmation Layers

For high-impact or sensitive governance actions — such as protocol upgrades, treasury disbursements, or DAO mergers—DAOs can introduce **mandatory human ratification**

steps. Even if agents propose or vote on such actions, execution would require a secondary layer of human approval, acting as a democratic circuit breaker.

3. Agent Identity Verification and Staking Mechanisms

To reduce the risk of sybil attacks and malicious coordination, agents could be **linked to unique on-chain identities, attestations, or reputation scores**. Staking mechanisms — where agents or their operators post collateral — can further align incentives, introducing economic penalties for harmful or dishonest behavior. This mirrors existing DAO practices like slashing or voting escrow (ve-token models), but applied to autonomous agents.

These governance challenges are not purely technical — they strike at the heart of **legitimacy** in decentralized systems. If autonomous agents can subvert collective decision-making, DeFAI protocols risk replicating the same centralization and opacity issues they aim to disrupt.

As **agent marketplaces and monetization frameworks accelerate**, governance could become one of the most critical domains for risk management and innovation. The ability to **balance agent autonomy with collective trust** will be a defining factor in the long-term success of AI-integrated financial networks.

Ultimately, the evolution of DeFAI governance will require more than just smart agents — it will demand **smarter systems** around them: robust safeguards, incentive-aligned frameworks, and adaptive governance structures that evolve alongside the agents themselves.

8/ Closing Thoughts

The rise of DeFAI — decentralized financial artificial intelligence — marks one of the most important inflection points in the evolution of crypto infrastructure. It introduces **an era where economic activity is increasingly automated, optimized, and influenced by intelligent agents** acting across chains, protocols, and governance layers.

From composable frameworks and scalable deployment protocols to autonomous agents and on-chain marketplaces, a full-stack ecosystem is emerging to support the industrialization of intelligent financial coordination.

But with this newfound power comes a deeper need for caution, design discipline, and open standards. **The same agents that enhance composability and performance could just as easily introduce new forms of centralization, opacity, or market instability if left unchecked.**

The future of DeFi is no longer purely human. It is modular, adaptive, and increasingly machine-mediated. Ensuring that this future remains transparent, resilient, and equitable will require proactive investment in security infrastructure, agent accountability, and governance innovation.

DeFAI is not just a technical upgrade — it is a new operating system for decentralized finance. What we choose to encode into it today will define the systems we inherit tomorrow.

9/ Appendix

Appendix 1

Prompt used to score Frameworks on the Autonomy-Specialization scale:

You are tasked with evaluating AI Agent Frameworks based on two key metrics: Level of Autonomy and Level of DeFi-Specialization. Use the following criteria for consistent scoring:

- **Level of Autonomy (1-5)**
 - 5: Provides comprehensive tools for building fully autonomous agents that function independently.
 - 4: Supports highly autonomous agents but requires some developer guidance.
 - 3: Facilitates semi-autonomous agents with flexible task configurations.
 - 2: Primarily serves as a manual toolkit, allowing for limited automation.
 - 1: Functions mostly as a static library or set of scripts with minimal automation.
- **Level of DeFi-Specialization (1-5)**
 - 5: Specifically designed to build agents focused on DeFi operations, including trading, liquidity management, and yield optimization.
 - 4: Primarily DeFi-focused, but also suitable for other financial automation.
 - 3: Supports both financial and non-financial agent development.
 - 2: Primarily non-financial, but has modular support for DeFi applications.
 - 1: General-purpose, with limited or no DeFi utility.

Evaluate the following AI Agent Frameworks based on current functionalities (not future plans): ElizaOS, ARC, GAME, Fetch.ai, Autonolas. Present your results in a structured table, and if scores differ from previous assessments, provide a rationale for the changes.

Appendix 2

Prompt used to score Agent Protocols on the Autonomy-Specialization scale:

You are tasked with assessing and ranking AI Agent Protocols based on two metrics: Level of Autonomy and Level of DeFi-Specialization. Use the following criteria for scoring:

- **Level of Autonomy (1-5)**

- 5: Fully autonomous, can execute end-to-end tasks independently.
- 4: Highly autonomous, capable of making strategic decisions with minimal human intervention.
- 3: Moderate autonomy, performs predefined tasks and adapts within set parameters.
- 2: Low autonomy, heavily dependent on human guidance for execution.
- 1: Minimal autonomy, primarily a manual or static tool.

- **Level of DeFi-Specialization (1-5)**

- 5: Exclusively designed for DeFi operations, such as trading, liquidity management, or yield optimization.
- 4: Primarily focused on DeFi but has potential applications outside financial contexts.
- 3: Balanced between DeFi and other use cases (e.g., social coordination, data processing).
- 2: Limited DeFi capabilities, designed mainly for non-financial applications.
- 1: Almost entirely non-DeFi oriented.

Evaluate the following AI Agent Protocols based on current functionalities (not future plans): Cod3x, Wayfinder, Modius, HeyAnon. Present your results in a structured table, and if scores differ from previous assessments, provide a rationale for the changes.

Appendix 3

Prompt used to score AI Agents on the Autonomy-Specialization scale:

You are tasked with assessing and ranking AI agents in the DeFAI space based on two metrics: Level of Autonomy and Level of DeFi-Specialization. Use the following criteria for scoring:

- **Level of Autonomy (1-5)**
 - 5: Fully autonomous, performs complex tasks independently.
 - 4: Highly autonomous, requires occasional human input.
 - 3: Moderate autonomy, performs predefined tasks with some adaptability.
 - 2: Low autonomy, relies on human guidance for most actions.
 - 1: Minimal autonomy, functions as a tool rather than an independent agent.
- **Level of DeFi-Specialization (1-5)**
 - 5: Exclusively DeFi-focused, handles financial operations independently.
 - 4: Primarily DeFi-focused, but can extend to related non-financial tasks.
 - 3: Balanced between DeFi and non-DeFi functions.
 - 2: Limited DeFi integration, primarily used for non-financial tasks.
 - 1: Non-DeFi focused, with minimal or no financial capabilities.

Assess the following agents based on current functionalities (not future capabilities): AIXBT, Griffain, Hive, SwarmNode, Orbit. Present your results in a structured table, and if scores differ from previous assessments, provide a rationale for the changes.

10 / References

1. <https://www.cookie.fun/>
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11 / New Binance Research Reports

Monthly Market Insights - May 2025 [Link](#)

A summary of the most important market developments, interesting charts and upcoming events



Pectra and Fusaka Upgrades: What does it mean for Ethereum? [Link](#)

Breaking down the upcoming Pectra and Fusaka upgrades and what it means for Ethereum in the long term



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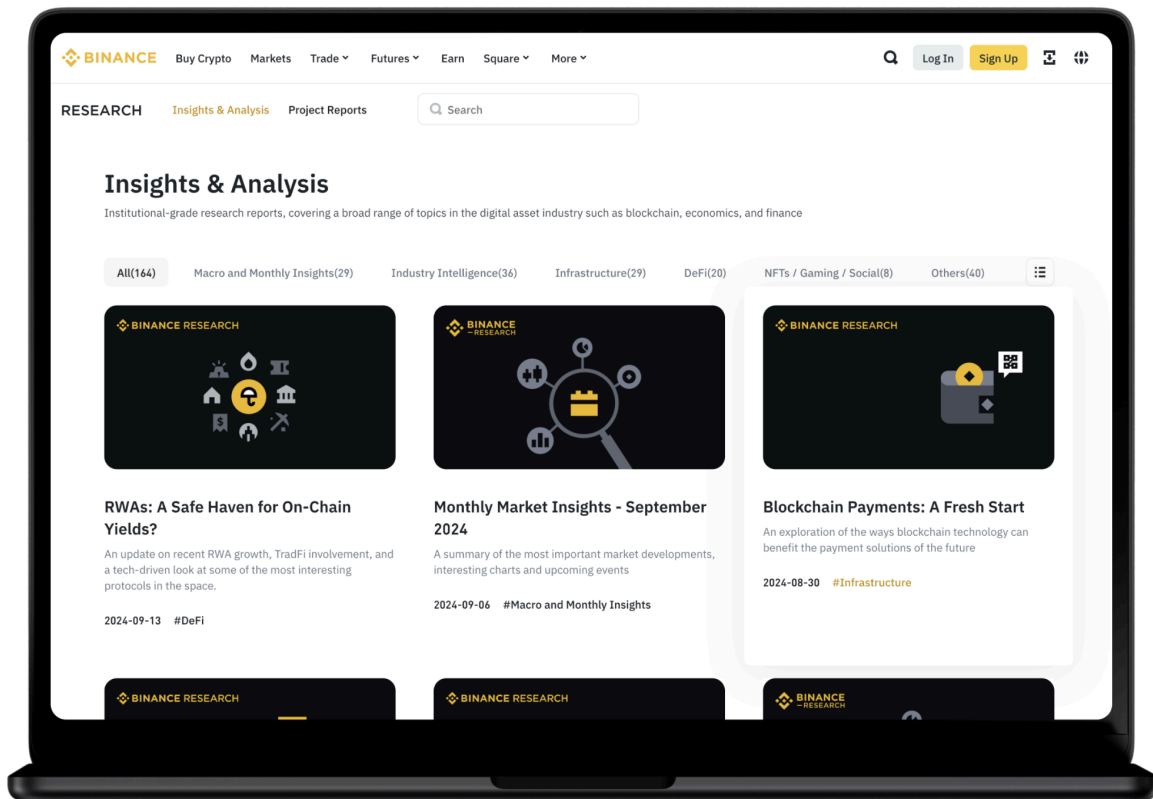


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